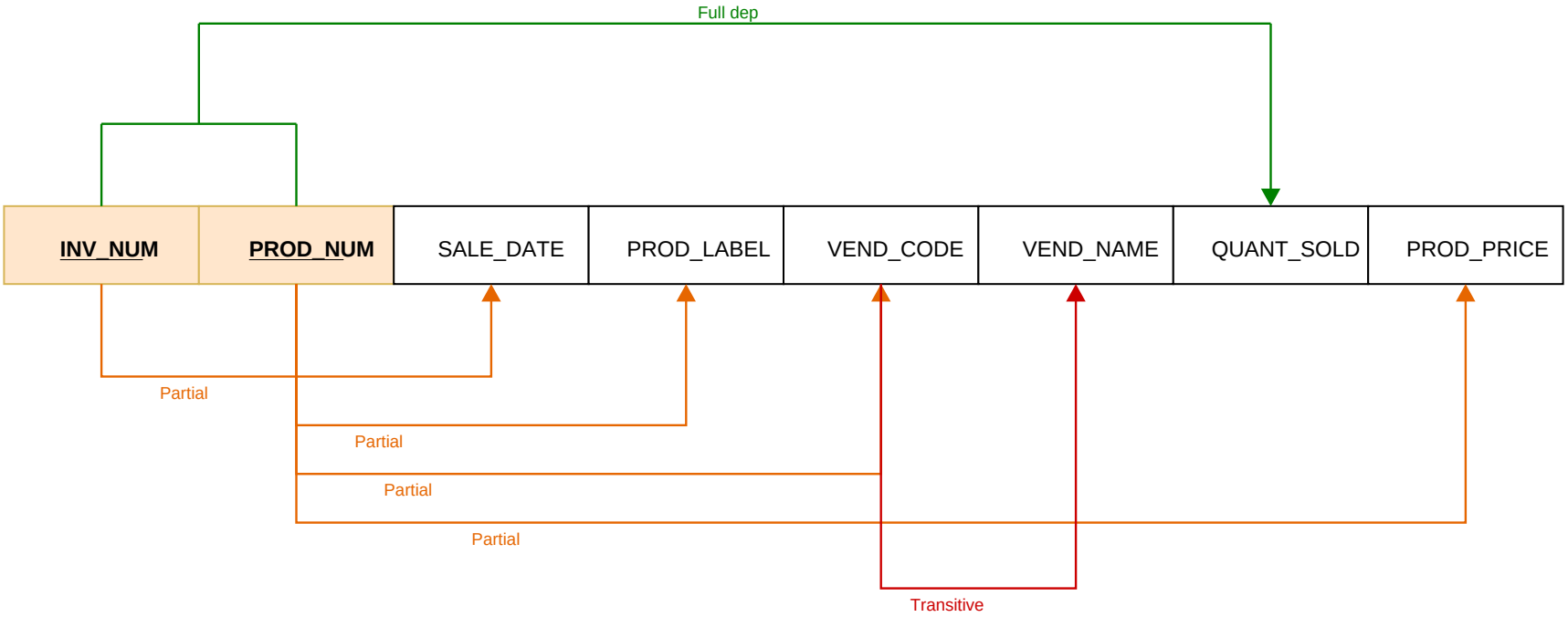
Q5 - INVOICE Table Dependency Diagram (1NF)

Orange+underline=PK Arrow above=Full dep Arrow below=Partial dep Arrow below=Transitive



Full: (INV_NUM, PROD_NUM) -> QUANT_SOLD

Partial: INV_NUM -> SALE_DATE | PROD_NUM -> PROD_LABEL, VEND_CODE, PROD_PRICE

Transitive: VEND_CODE -> VEND_NAME

Q6 - Dependency Diagrams After Removing Partial Dependencies (2NF)

Partial deps resolved. 3 tables created. Transitive dep still present in PRODUCT.

Table: INVOICE

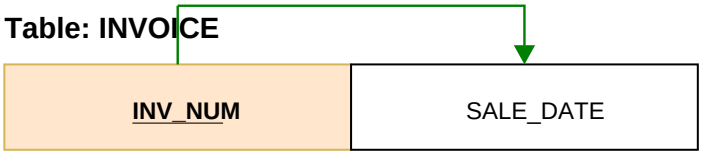


Table: PRODUCT

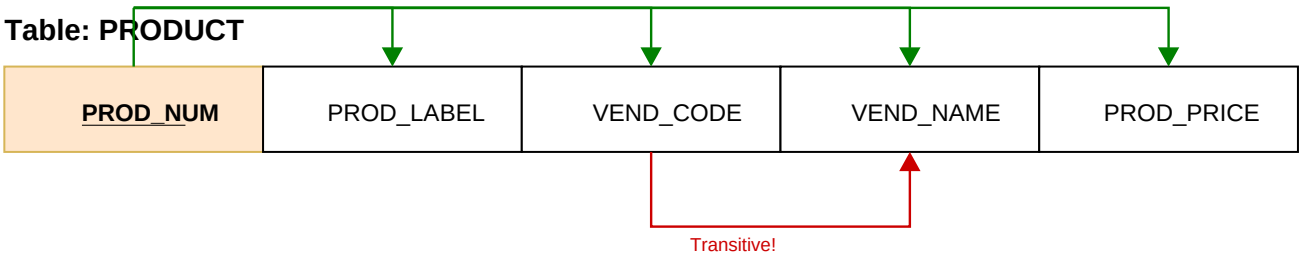
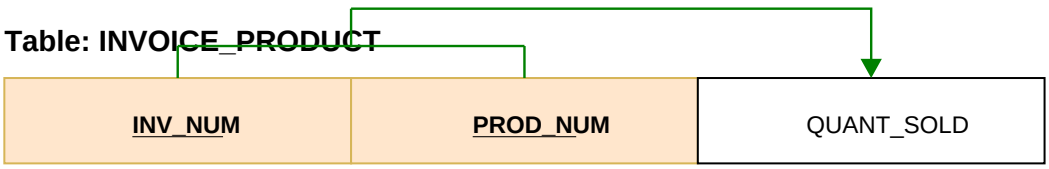


Table: INVOICE_PRODUCT



2NF Result (3 tables):

INVOICE (INV_NUM, SALE_DATE)

PRODUCT (PROD_NUM, PROD_LABEL, VEND_CODE, VEND_NAME, PROD_PRICE)

INVOICE_PRODUCT (INV_NUM, PROD_NUM, QUANT_SOLD)

Transitive dep VEND_CODE->VEND_NAME still in PRODUCT -- resolved in Q7

Q7 - Dependency Diagrams After Removing Transitive Dependencies (3NF)

VEND_CODE->VEND_NAME extracted to new VENDOR table. Database now fully in 3NF.

Table: INVOICE

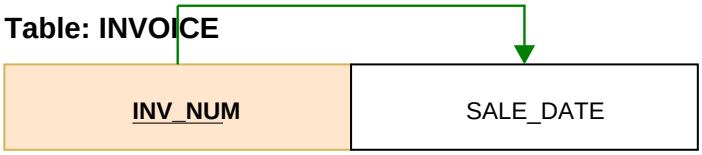


Table: PRODUCT

VENDOR (new)

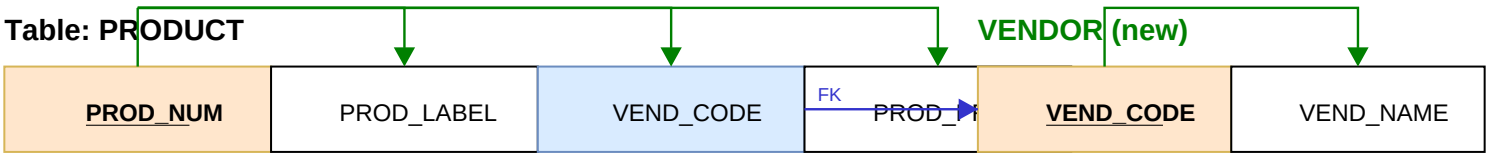
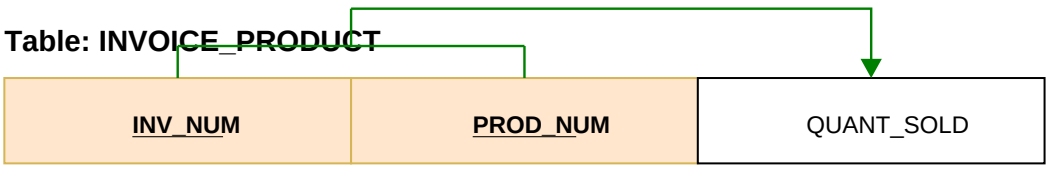


Table: INVOICE_PRODUCT



3NF Final Result (4 tables):

INVOICE (INV_NUM, SALE_DATE)

PRODUCT (PROD_NUM, PROD_LABEL, VEND_CODE [FK], PROD_PRICE)

VENDOR (VEND_CODE, VEND_NAME)

INVOICE_PRODUCT (INV_NUM, PROD_NUM, QUANT_SOLD)

No partial deps. No transitive deps. Each table = single subject. Fully normalised.